UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

Unfolding

Lecture 19

Course of:

Statistics and data analysis

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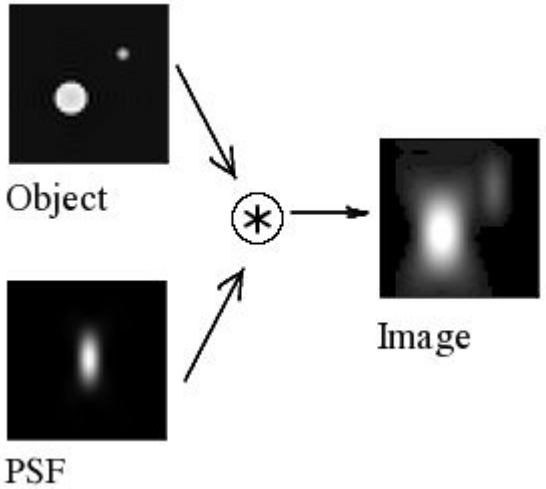
Unfolding

- Cowan, cap. 11

Context

- When acquiring data with a finite resolution detector, observables are smeared. The **true distribution of the physical observables is not accessible**
  - Effect of imperfect detector acceptance
  - Loss of events due to trigger inefficiencies
  - Background processes interpreted as signal
  - …

- The goal is to **unfold** the response of the detector and **access the original information**



The **need for unfolding**

- Unfolded data allows for immediate, direct **comparison between the experiment results and the theoretical model**
- Unfolded data is the **fair way to compare results from different experiments**

- In general it is an attempt to access the measurement of the **pure physics observables** without any smearing due to the instrumentation

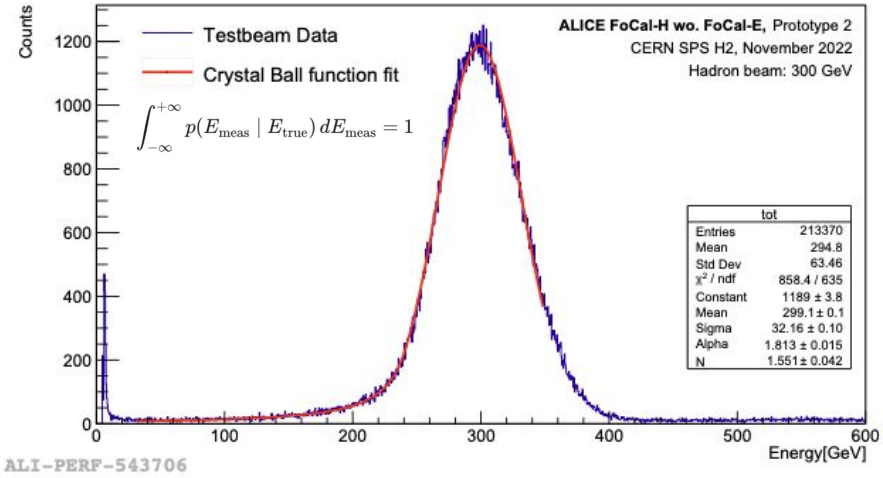
- **⇒ Statistical challenge of <u>reconstructing the cause, starting from the effect</u>**

The **response function**

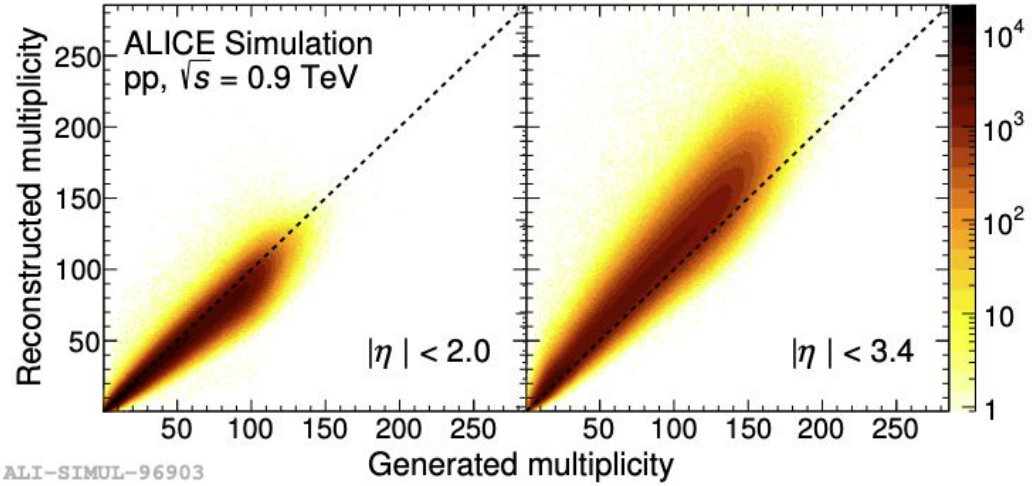
- The response function describes the **distribution of the measured value for a given true value**

$$R(E_{\text{meas}} \mid E_{\text{true}}) = p(E_{\text{meas}} \mid E_{\text{true}})$$

- In the simplest case the smearing is described by a gaussian
  - Usually the sum of many different effects
- When plotting generated vs. reconstructed, **need to normalise (pdf)**
- <u>From now on, the assumption is that **we know the response of the detector perfectly**</u>
  - ■ This is not always true in the real world



ALICE Collaboration



The **response function**

- When considering continuous variables:

  $f_t(x_t)$: distribution of true values (normalized to unity)

  $f_m(x_m)$: distribution of measured values (normalized to unity)

  $f_b(x_m)$: distribution of background (normalized to unity)

- Response function R:

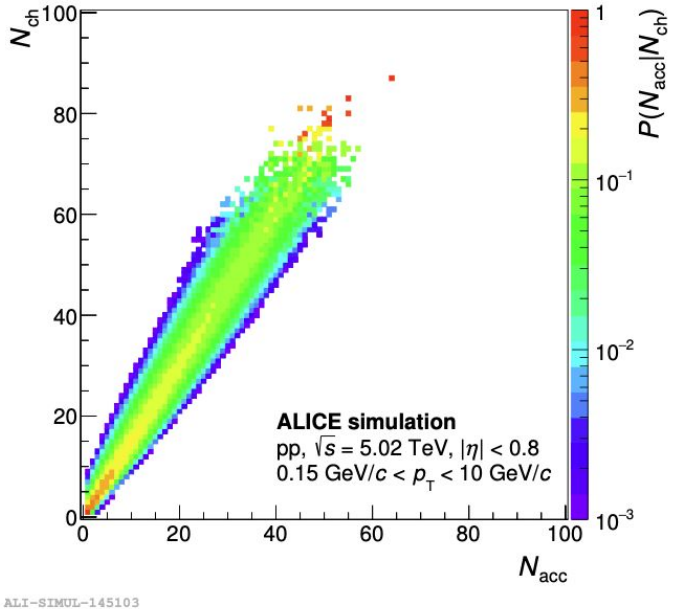
$$R(x_m|x_t) = r(x_m|x_t) \times \varepsilon(x_t)$$

probability (density) to observe x_m given x_t

"smearing" "efficiency"

- If we consider the full measurement space:

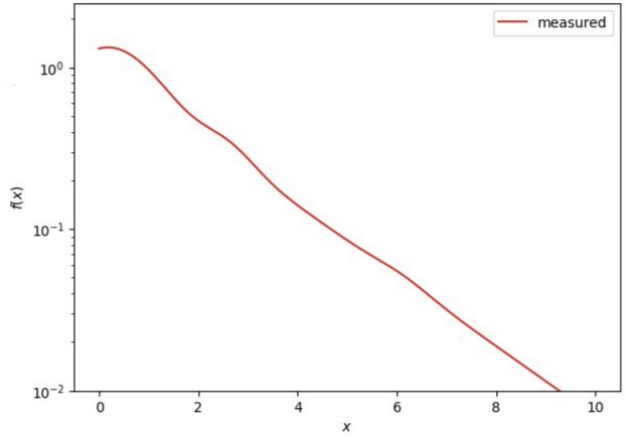
$$\int_{\Omega_m} r(x_m|x_t)\,\mathrm{d}x_m = 1$$



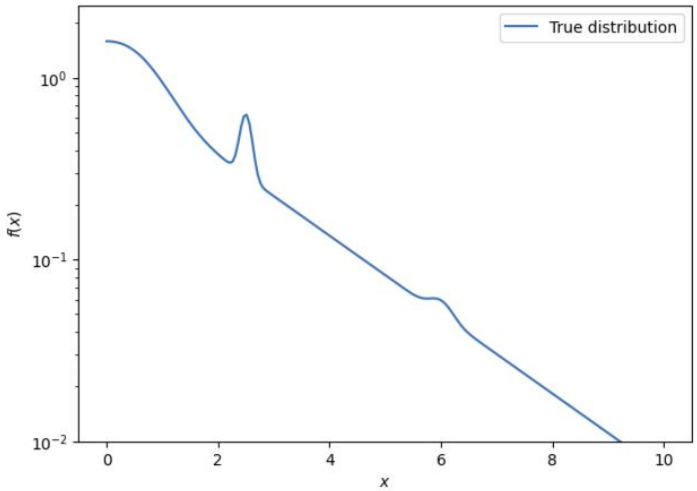
Comparison of created and accepted charged particles

In short

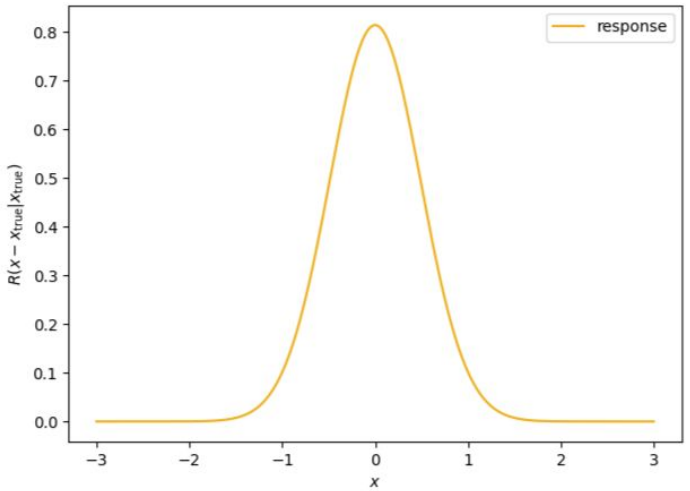
- For fixed response the effect is a convolution
- Can we undo the convolution somehow (“**deconvolution**”)?
- What about the uncertainties?



?

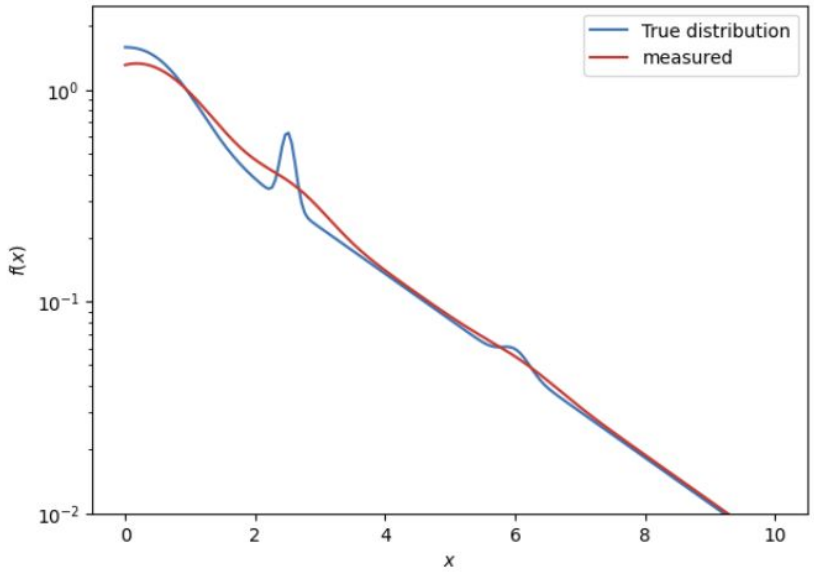


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Observations

- The detector response tends to **“wash out” peaks and other details**
- Steeper parts of the distribution become less steep
- **Some information is lost due to the smearing**
- Intuition: this should lead to **large uncertainties when inverting**
- Details smaller than the detector resolution should be hard to detect



The **discretized approach**

- The Response Matrix is the **discrete version of the response function**
  - Ideally it **encapsulates all detector effects into a single entity**
  - It represents the conditional probability P(y|x) → probability of measuring a value y given that the true physical value was x
  - The response function intrinsically includes finite resolution, geometric acceptance limitations, and efficiencies
  - The discretization is a mathematical necessity
    - Continuous folding is described by $g(y) = \int K(y,x) f(x)\, dx$ where K is a compact operator → its inverse isn't continuous

m_{tot} : total number of true events

n_{tot} : total number of measured events

b_{tot} : total number of background events

$$\mu_{\text{tot}} = E[m_{\text{tot}}], \quad \nu_{\text{tot}} = E[n_{\text{tot}}], \quad \beta_{\text{tot}} = E[b_{\text{tot}}]$$

Probability to find x_t in bin j → **discretization**

$$p_j = \int_{\text{bin } j} \mathrm{d}x_t \, f_t(x_t), \quad \mu_j = \mu_{\text{tot}} \times p_j$$

Expected measured number of entries in bin *i*:

$$\nu_i = \mu_{\text{tot}} \int_{\Omega_t} \mathrm{d}x_t \, \text{Prob}(x_m \text{ in } i | \text{true } x_t, \text{ detected})$$
$$\times \text{Prob}(\text{detect } x_t) \times \text{Prob}(\text{produce } x_t)$$
$$= \mu_{\text{tot}} \int_{\text{bin } i} \mathrm{d}x_m \int_{\Omega_t} \mathrm{d}x_t \, r(x_m|x_t)\varepsilon(x_t) f_t(x_t)$$

No background

True space

Building **response matrix**

Further definitions:

$$\int_{\Omega_t} \mathrm{d}x_t = \sum_{j=1}^{M} \int_{\text{bin } j} \mathrm{d}x_t$$

$$\nu_i = \mu_{\text{tot}} \int_{\text{bin } i} \mathrm{d}x_m \sum_{j=1}^{M} \int_{\text{bin } j} \mathrm{d}x_t \, r(x_m|x_t)\varepsilon(x_t)f_t(x_t)$$

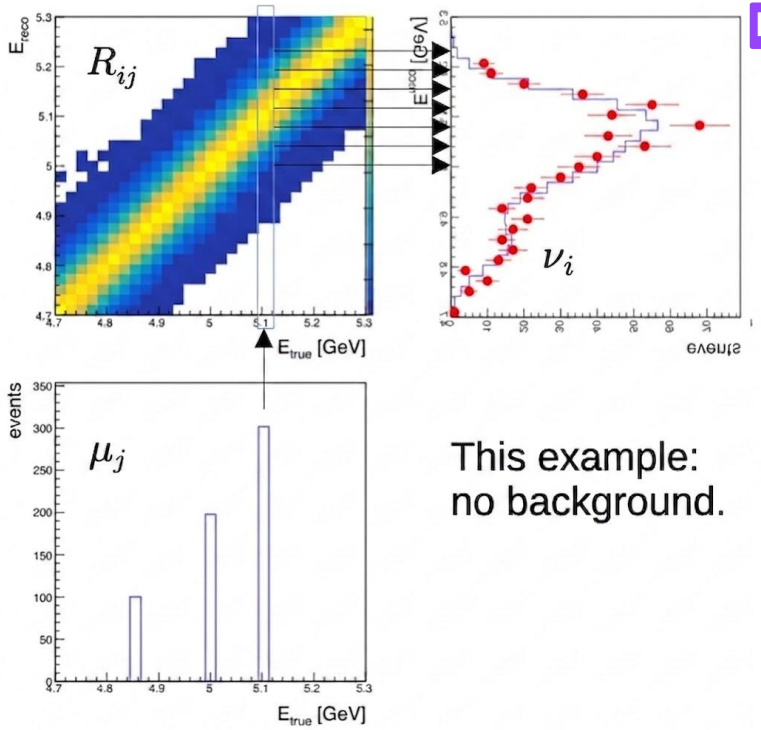
$$= \sum_{j=1}^{M} \int_{\text{bin } i} dx_m \int_{\text{bin } j} \mathrm{d}x_t \, \frac{r(x_m|x_t)\varepsilon(x_t)f_t(x_t)}{\mu_j/\mu_{\text{tot}}} \mu_j$$

This may be written as

$$\nu_i = \sum_{j=1}^{M} R_{ij}\mu_j$$

with the components of the response matrix R_{ij} given by

$$R_{ij} = \frac{\int_{\text{bin } i} dx_m \int_{\text{bin } j} \mathrm{d}x_t \, r(x_m|x_t)\varepsilon(x_t)f_t(x_t)}{\int_{\text{bin } j} \mathrm{d}x_t f(x_t)}$$



This example:
no background.

Element R_{ij} is the probability that **an event generated in true bin j is reconstructed in measured bin i**

The discrete **folding equation**

- In short

$$R_{ij} = \text{Prob}(\text{observed in bin } i | \text{true in bin } j)$$

- Summing the response matrix over i gives the efficiency:

$$\sum_{i=1}^{N} R_{ij} = \varepsilon_j$$

- In matrix form and including background:

$$\nu_i = \sum_{j=1}^{M} R_{ij}\mu_j + \beta_i$$

$$\vec{\nu} = R\vec{\mu} + \vec{\beta}$$

This is the folding equation

- Response matrix depends on $f_t(x_t)$ which we want to know. However, if we make the bins small enough $f_t(x_t) \approx$ const within a bin and drops from the ratio:

$$R_{ij} = \frac{\int_{\text{bin } i} dx_m \int_{\text{bin } j} \mathrm{d}x_t \, r(x_m|x_t)\varepsilon(x_t)f_t(x_t)}{\int_{\text{bin } j} \mathrm{d}x_t f(x_t)} \approx \frac{1}{\Delta x_{t,j}} \int_{\text{bin } i} dx_m \int_{\text{bin } j} \mathrm{d}x_t \, r(x_m|x_t)\varepsilon(x_t)$$

Considerations on the response matrix

- The response matrix usually comes from simulations. **The assumption is that it is perfectly known**
  - Use MC to build the true vs reconstructed matrix
  - Need to simulate 2 ingredients: detector response + physics of the phenomenon (true distr)

- If true distribution is different from simulated, **this can give a bias**
  - Distorted estimate of events migration from one bin to another due to wrong modeling of the true distr

- That is only a problem, if the response varies strongly over a bin
  - Bias is 0 if the event distribution can be considered flat within the bin

- **Smaller bins are better in this regard → less exposed to potential bias**

The **inverse matrix** approach

Unfolding by **inverting the response matrix**

- We have

$$\vec{\nu} = R\vec{\mu} + \vec{\beta} \qquad \rightarrow \textbf{Folding equation (linear!)}$$

- Replace $\vec{\nu}$ by $\vec{n}$ to obtain an obvious estimator for the true distribution:

$$\hat{\vec{\mu}} = R^{-1}(\vec{n} - \vec{\beta})$$

- It can be shown that this solution minimizes

$$\chi^2(\vec{\mu}) = \underbrace{(\vec{\nu}(\vec{\mu}) - \vec{n})}^{\mathsf{T}} V^{-1} \underbrace{(\vec{\nu}(\vec{\mu}) - \vec{n})} \qquad \text{where} \qquad V_{i,j} = \mathrm{cov}[n_i, n_j]$$

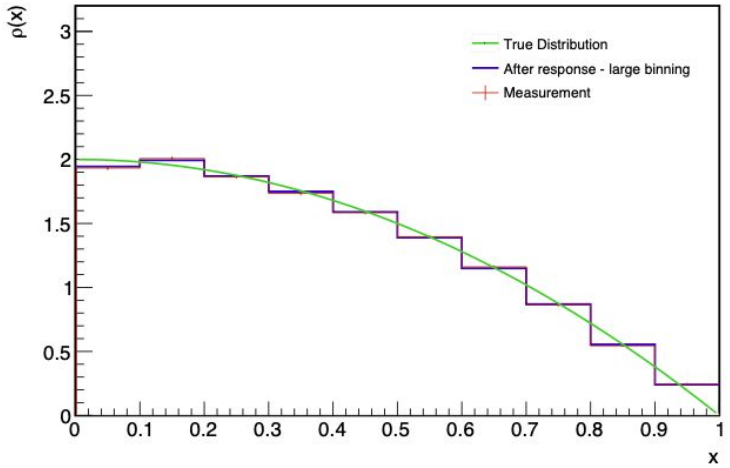
Measures the distance, bin-by-bin, between the data and the projection of the true distribution

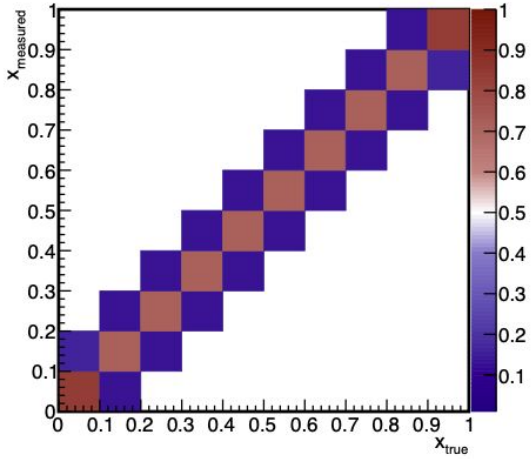
- with the covariance matrix of the solution is given by

$$U = R^{-1} V (R^{-1})^{\mathsf{T}}$$

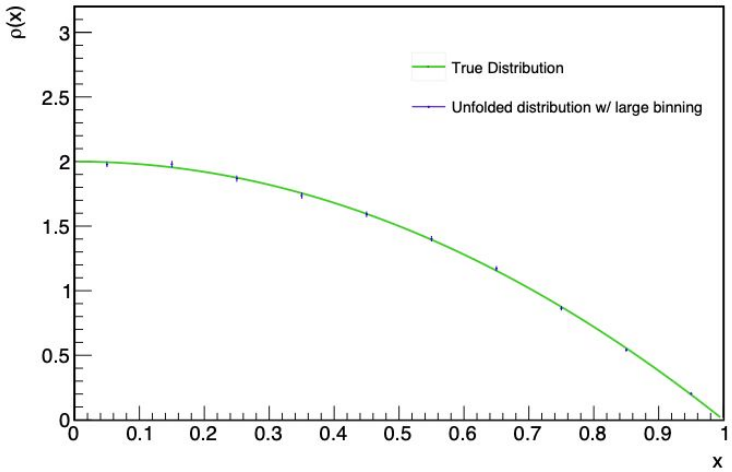
- This is the analytical solution to the original problem!
  - Equivalent to set the distance between the folded true distribution and the data to zero
  - → **analytical overfitting** by construction! (fit N points with a function with N parameters)

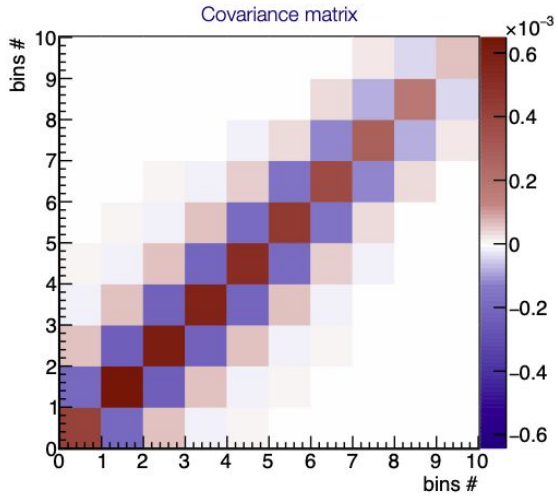
Example - **large bins** compared to resolution





- Response is Gaussian, but with a **width less than one bin** (σ = 0.055)
- Measurement fluctuates around the convoluted distribution

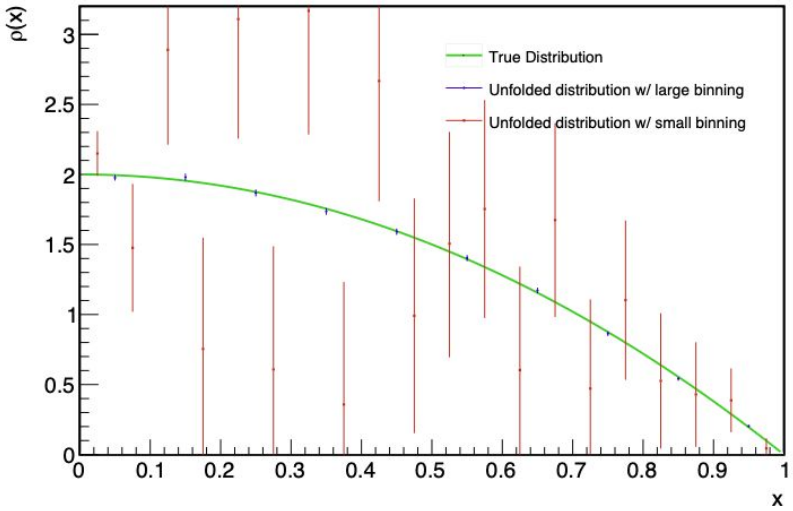




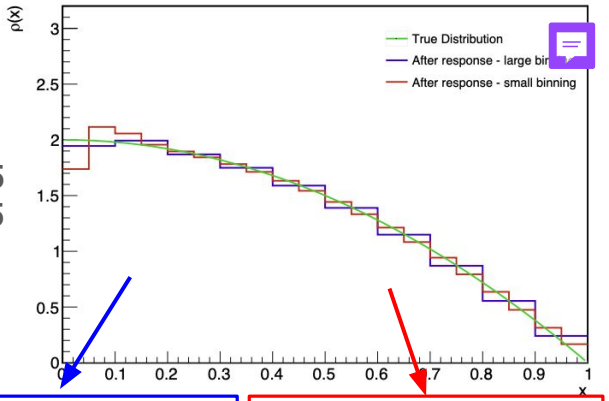
- Unfolded distribution agrees with true distribution - **deconvolution is successful**
- In this case, all counts can come from current or adjacent bin
  - The reconstruction of any given bin is deeply entangled with its adjacent bins
  - When the unfolding algorithm assigns a fluctuation to a specific bin, it must mathematically subtract it from its neighbors to conserve the total number of observed counts
  - → **covariance matrix shows anticorrelation between adjacent bins**

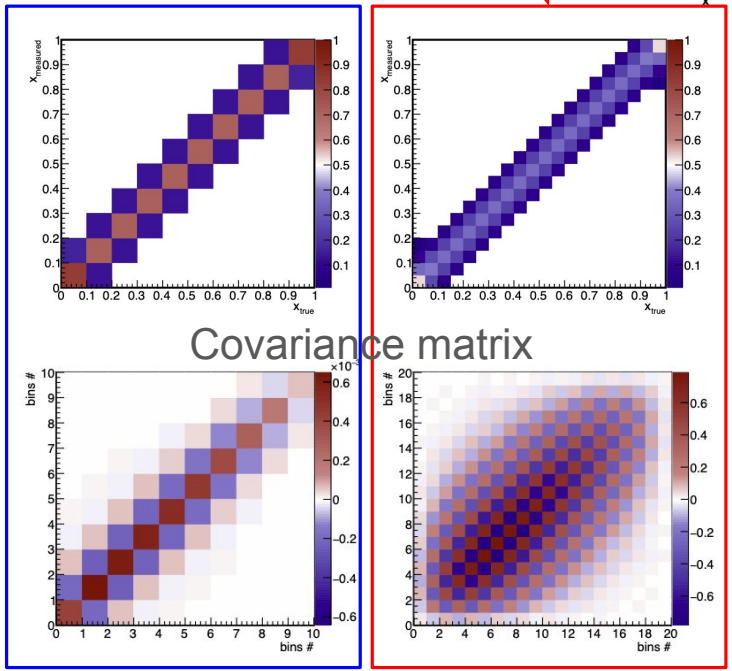
Example - **small bins** compared to resolution

Unfolded result



Large bins: 0.1
Small bins: 0.05
Response: σ = 0.055





- **Unfolded distribution very different from the true one for the small bins case**
  - Large fluctuations + crazy error bars
- Shouldn't the small bins case be better?

What is the problem when inverting the matrix?

- Applying the response matrix R smears out fine structure
- **Applying R^{-1} creates unwanted structures**
  - Statistical fluctuations are treated as physical signals
  - Result in bin-by-bin artificial compensations
  - Unfolding is mathematically classified as an ill-posed inverse problem.
  - The solution lacks continuous dependence on the data. Small input changes cause unbounded output changes.

- More desirable solution by adding (smoothness) constraints. However, this will produce a bias.
- Depending on use case, find an acceptable balance between bias and smoothness

- Bias vs Variance trade off:
  - **Large bins → stat fluctuations are less important → stable behaviour, but bias**
  - **Small bins → small bias but increased noise → unstable behaviour** (analytical overfitting)

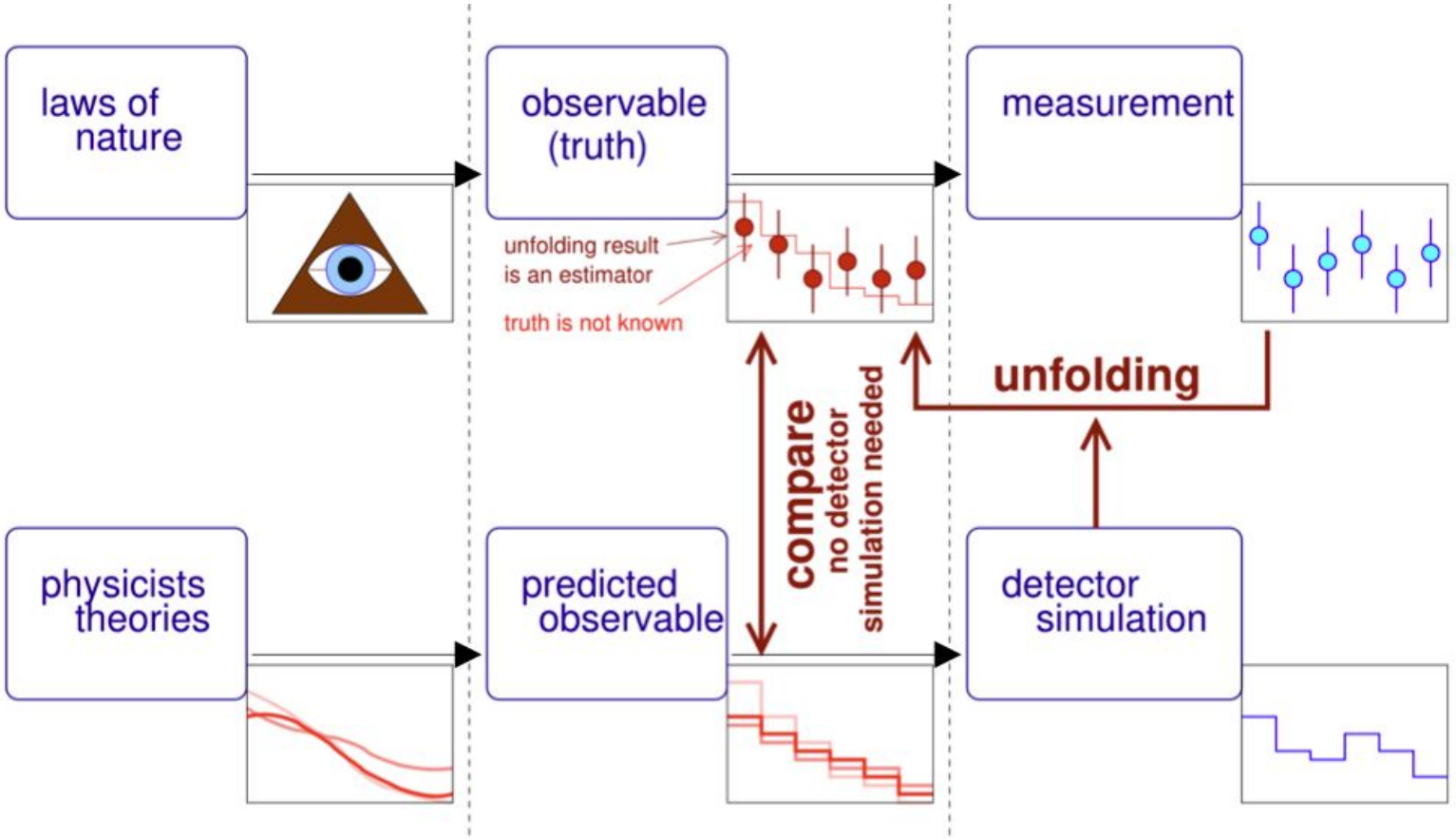
- → **Regularization methods** introduced to control the oscillations and make a small bins approach viable

To unfold or not to unfold?

- “**The most important advice I can give about deconvolution is: Don't**. It's a lot of work, and often produces biased or otherwise unsatisfactory results. Moreover it's often unnecessary.”

  Oser’s lecture

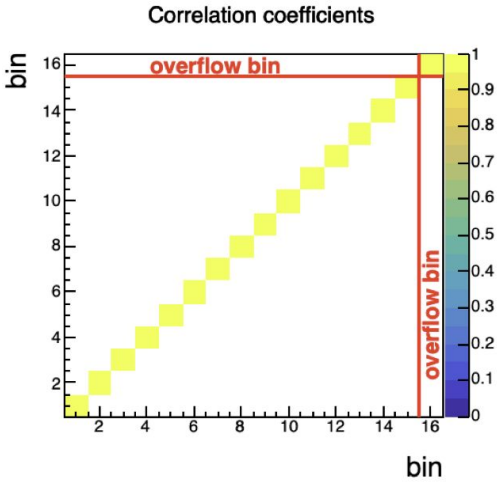
- "**Forward folding**" is much easier:
  - Take theory prediction
  - Convolve it with the response of the detector
  - **Compare smeared theory directly with the data**
    - Chi2 or Likelihood fit


laws of nature
observable (truth)
measurement
unfolding result is an estimator
truth is not known
unfolding
compare
no detector simulation needed
physicists theories
predicted observable
detector simulation

Bin-by-bin method

Goal: **avoid the matrix inversion**

- This method intentionally **avoids matrix inversion entirely**
- It makes the two strong assumptions:
  - **Bin migration effects** (off-diagonal elements) **are negligible**
    - Instead of a matrix operation, it applies a simple multiplicative correction factor to each bin independently
  - **Monte Carlo physics shape matches the underlying physical data shape perfectly** within each bin



- Determine correction factor for each bin on MC by computing:

$$\mu_i = C_i(n_i - \beta_i) \qquad C_i = \frac{\mu_i^{\mathrm{MC}}}{\nu_i^{\mathrm{MC}}}$$

- Expectation value of the corrected data:

$$E[\hat{\mu}_i] = C_i E[n_i - \beta_i] = C_i(\nu_i - \beta_i) \equiv C_i \nu_i^{\text{sig}}$$

- By inserting the C_i one can determine the bias:

$$E[\hat{\mu}_i] = \frac{\mu_i^{\text{MC}}}{\nu_i^{\text{MC}}} \nu_i^{\text{sig}} = \underbrace{\left(\frac{\mu_i^{\text{MC}}}{\nu_i^{\text{MC}}} - \frac{\mu_i}{\nu_i^{\text{sig}}} \right) \nu_i^{\text{sig}}}_{\text{bias}} + \mu_i$$

no bias only if MC = nature

- Covariance matrix of the corrected data (smearing fluctuations independent between bins)

$$U_{ij} = \text{cov}[\hat{\mu}_i, \hat{\mu}_j] = C_i C_j \underbrace{\text{cov}[n_i^{\text{sig}}, n_j^{\text{sig}}]}_{0 \text{ for } i \neq j} = C_i^2 \text{Var}[n_i^{\text{sig}}] \delta_{ij}$$

When the bin-by-bin approach can be used

- **Broad bins:** when chosen bin widths are much larger than the detector resolution, naturally minimizing migration
- **Low migration:** when the response matrix is highly diagonal (Purity > 70-80% in all bins)
- **Early feedback:** useful for quick, robust first looks at data before deploying complex algorithms

Regularized unfolding methods

Maximum likelihood formulation and general strategy for regularization

- Matrix inversion is the maximum likelihood solution:

Independent Poisson fluctuations: $\ln L(\vec{\mu}) = \sum_{i=1}^{M} (n_i \ln \nu_i - \nu_i)$

ML estimator: $\hat{\vec{\nu}} = \vec{n}$

$\rightarrow \hat{\vec{\mu}} = R^{-1}(\vec{n} - \vec{\beta})$

- Idea behind the regularization: **accept solutions which are close to the maximum likelihood estimate**
  - Arbitrary choice for the tolerance band
  - Within the band pick the **most regular solution**

E.g. 1σ band

$$\ln L(\vec{\mu}) \geq \ln L(\vec{\mu}_{\text{max}}) - \Delta \ln L(\vec{\mu})$$

- Introduce a **smoothness function S** so that the re-defined task is to maximize:

$$\phi(\vec{\mu}) = \ln L(\vec{\mu}) + \tau S(\mu)$$

τ is a tunable parameter and controls the bias-variance tradeoff

Tikhonov regularization

Smoothness measure for the deconvoluted function *f*:

$$S[f] = -\int \mathrm{d}x \left(\frac{\mathrm{d}^k f}{\mathrm{d}^k x}\right)^2 \qquad k = 1, 2, 3, \ldots$$

Minus sign makes *S* small when *k*-th derivative is large

Tikhonov for $k = 2$:

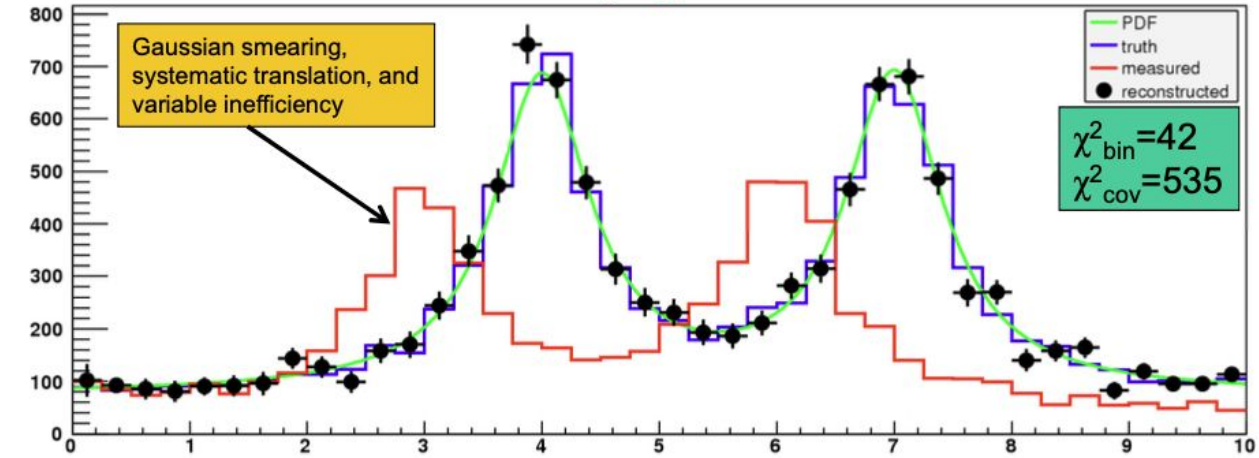
numerical estimate of the second derivative:

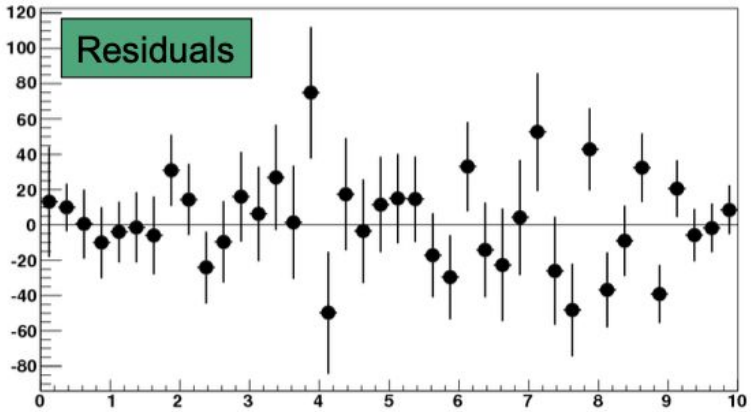
$$\phi(\vec{\mu}) = \ln L(\vec{\mu}) + \tau S(\mu), \qquad S(\vec{\mu}) = -\sum_{i=1}^{M-2} (-\mu_i + 2\mu_{i+1} - \mu_{i+2})^2$$

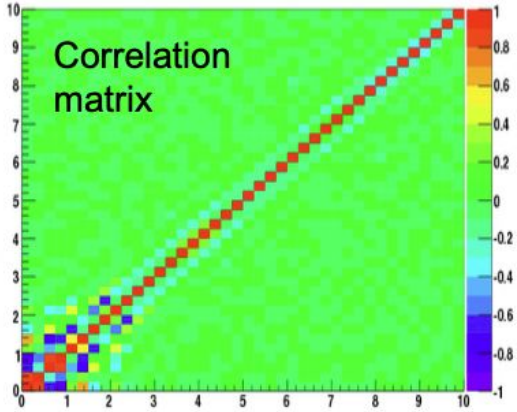
Implementation by A. Höcker, V. Kartvelishvili: *Singular Value Decomposition*
(NIM A372 (1996) 469, hep-ph/9509307, TSVDUnfold in ROOT)

Minimizes $\chi^2(\vec{\mu}) + \tau \sum_i [(\mu_{i+1} - \mu_i) - (\mu_i - \mu_{i-1})]^2$

- In practice it introduces a **penalty term for the changes in the slope**
- To be interpreted as a prior information on the smoothness of the result







The choice of k defines the **geometrical trend we define as smoothness criterion**

For example:
k = 1 ⟶ regularization of steepness
k = 2 ⟶ regularization of curvature
…

Typically k=2 is a good choice

http://hepunx.rl.ac.uk/~adye/software/unfold/RooUnfold.html
Tim Adye, Unfolding in HEP, https://indico.cern.ch/event/671301/contributions/2745801

Iterative unfolding
(D'Agostini method)

The **forward folding** approach

- Compute the **probability for a true signal to generate the measured effect**
  - Use the Bayes theorem applied to our discrete distributions

$$P(C_j|E_i) = \frac{P(E_i|C_j) \cdot P(C_j)}{\sum_k P(E_i|C_k) \cdot P(C_k)}$$

C_j = true event in bin j = **cause**
E_i = measured event in bin i = **effect**

- $P(E_i | C_j)$ = this is the **response matrix** usually from MC simulations
- $P(C_j)$ = prior on the true distribution, **initial hypothesis on the phenomenon**
- $P(C_j | E_i)$ = probability that the **effect seen in bin *i* is caused by the true phenomenon in bin *j***

From **conditioned probability** to the **number of events**

- Once we compute the posterior $P(C_j \mid E_i)$ we can use it to compute the total **number of true events bin-by-bin**:

$$\hat{n}(C_j) = \frac{1}{\epsilon_j} \sum_{i=1}^{M} n(E_i) \cdot P(C_j|E_i)$$

$n(E_i)$ = number of events measured in bin i
ϵ_j = efficiency of the detector for true bin j

- Intuitively:
  - Take each event measured in bin i
  - Sort it back to the true bin j
  - Weight it with the Bayesian probability $P(C_j \mid E_i)$

- **Observation**: to compute $n(C_i)$ we need to know $P(C_j \mid E_i)$, to compute $P(C_j \mid E_i)$ we need to know the prior, but $P(C_j)$ is the normalized version of the true distribution we are trying to compute!

Iterative approach to solve a circular argument

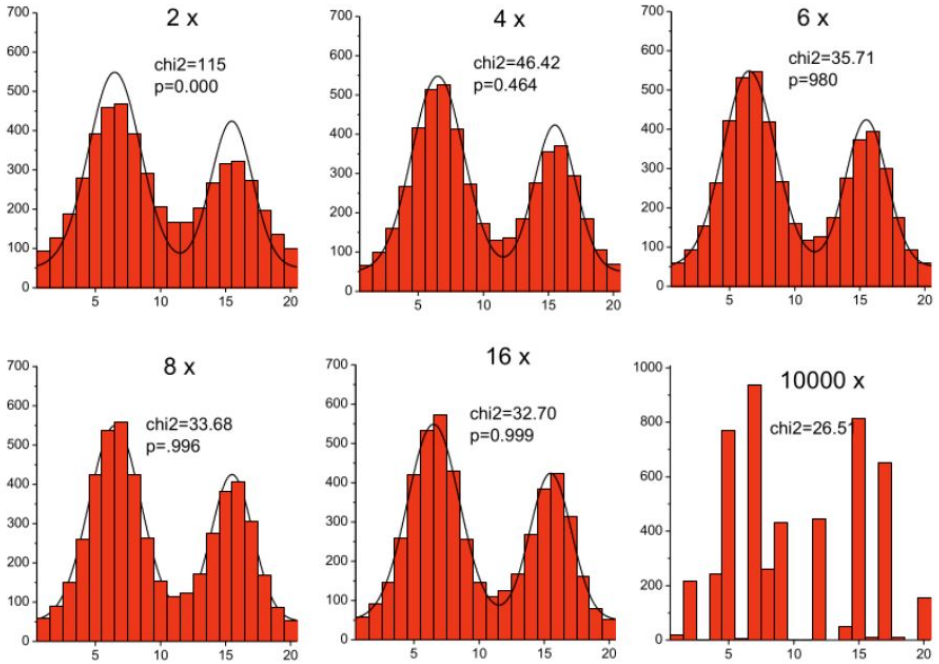
- **Start**: pick the initial prior $P_0(C_j)$
  - Usually, the theoretical Monte Carlo distribution, or a flat (uniform) distribution if no assumption is desired
- **Calculation Phase**: Using $P_0(C_j)$ and the detector matrix to compute the posterior $P(C_j \mid E_i)$
- **Unfolding Phase**: The posterior is applied to the real measured data $n(E_i)$ to obtain a first estimate of the unfolded spectrum $n_1(C_j)$
- **Update** (feedback): This new estimate $n_1(C_j)$ is transformed into the new prior for the next iteration: $P_1(C_j) = n_1(C_j)/\sum n_1$
- **Repeat**: The process restarts using $P_1(C_j)$ instead of $P_0(C_j)$. The cycle continues.

Stopping condition

- Limited number of iterations provides implicit regularization
  - **Early stopping!**

- Few iterations → the **initial prior dominates**
  - High bias
- Too many iterations → the algo forgets the prior → **large stats fluctuations**
  - High variance

- Typically 3-6 iterations are sufficient
- `PyUnfold` uses a Kolmogorov-Smirnov test to compute the distance between the distribution of consecutive iterations and decide when to stop



Summary

- Each measurement “folds” true distribution with response matrix. **Unfolding aims to invert this step**
- Sometimes it is just easier to publish the measured distribution + response

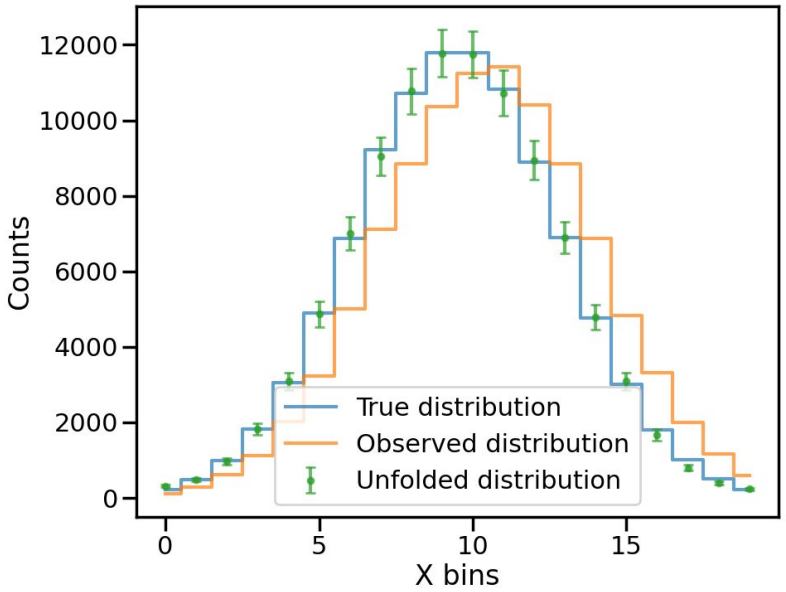
- If response is wide compared to binning, individual bins have large uncertainties and strong correlations
  → **regularization methods**

- **Regularisation** makes the result easier to interpret, but **introduces a bias**
  - Make sure the added information about smoothness is information you actually have

Examples…

Iterative unfolding with **PyUnfold**

- Here is an example of how to use the iterative Bayesian approach with PyUnfold
- The result is satisfactory in this case
  - Synthetic data are used



Exercises…

Implement the other methods and discuss the results

- Starting from the previous example:
  - Implement the inverse matrix method: when does it return acceptable results?
  - Implement the bin-by-bin method: when does it return acceptable results?

- Plot the unfolded distribution and the covariance matrix and discuss your findings in terms of accuracy, uncertainty, …